

Deep-research — Wiring the dormant pkg/verification multipass polisher into unified-translator

Revision: 1 **Last modified:** 2026-06-15T00:00:00Z **Author:** background feature-completion stream (§11.4.124 / §11.4.150) **Scope:** cmd/unified-translator/main.go + pkg/verification

1. The gap (FACT, this session)

pkg/verification contains a substantial, actively-maintained multi-pass LLM verification/polishing engine (MultiPassPolisher, BookPolisher, Verifier, note-taking, SQLite persistence, reporter). It is **never wired into any cmd/ binary**. The CLI `-verify` flag only runs lightweight heuristic output-quality checks (verifyTranslation = target-script character presence; verifyEPUB = structural check). The real multipass *polishing* capability ships dormant.

Git-history investigation (§11.4.124 — investigate-before-wire)

- `git log --all -S "pkg/verification" -- cmd/` → **empty**. No cmd/ binary has *ever* imported the package across the entire history.
- First add: commit 717bf18 (“Auto-commit”, 2025-11-20) created the package already unwired, alongside Documentation/VERIFICATION_SYSTEM.md.
- Current importers (whole tree): only test/performance/..., test/unit/verification_test.go, test/e2e/... plus the package’s own _test.go files. Zero production importers.
- verifier.go has 5 commits; multipass.go carries recent mutation-proven defensive fixes (chapter-count cap to avoid panic; high-water-mark dedup of DB writes; drop-empty-provider consensus to avoid score dilution / content-wipe). This is **production code that was finished but never connected**, not scaffolding.

Determination: genuinely-built capability, never wired (born unwired in an auto-commit). Per §11.4.124 the correct action is to **wire it in properly**, not remove it.

Engine soundness check (is it correct, or scaffolding?)

BookPolisher.polishSection → verifyWithLLM → translator.Translate (real LLM call) → parseVerificationResponse (parses SPIRIT/LANGUAGE/CONTEXT/VOCABULARY scores, ISSUES, POLISHED_TEXT) → buildConsensus (averages scores, picks polished candidate, **defaults to the**

existing translation and honours a **UNCHANGED** sentinel). The engine genuinely calls the LLM and applies a parsed, consensus-selected polished version — it is not a stub.

2. Deep multi-angle research (§11.4.150 — ≥ 2 angles, latest sources)

Angle 1 — Technique: iterative LLM MT post-editing is established practice

The canonical translation-specific method is **TEaR (Translate, Estimate, Refine)** — exactly the “translate → verify/critique → polish” shape we are wiring. Key findings:

- A **single** refinement round is best by design; COMET peaks at pass 1 then plateaus/declines (78.67 → 79.50 → 79.29 → 79.28 on WMT23 Zh-En), later passes regress due to hallucinated estimation feedback. <https://arxiv.org/html/2402.16379v3> (accessed 2026-06-15); <https://arxiv.org/abs/2402.16379> (accessed 2026-06-15).
- Stopping = a quality-estimation **gate**: if no errors detected, return the original **unchanged** — the built-in regression guard.
- Multi-LLM consensus per pass (M-MAD / multi-agent debate, best-of-N QE reranking) is a credible *verification*-stage pattern. <https://www.emergentmind.com/topics/multi-agent-debate-mad-paradigm> (accessed 2026-06-15); <https://arxiv.org/html/2510.08870v1> (accessed 2026-06-15).
- Standard gating metrics: COMET / GEMBA / LLM-as-judge. <https://kvashee.medium.com/mt-quality-evaluation-in-the-age-of-llm-mt-ca8a9540a2ab> (accessed 2026-06-15).

Angle 2 — Risk / failure-modes: refinement can **HARM** without guards

- Intrinsic self-correction without an external signal often fails to improve or **degrades** output (inherits the same hallucinations that caused the original error). TACL critical survey “When Can LLMs Actually Correct Their Own Mistakes”: https://direct.mit.edu/tacl/article/doi/10.1162/tacl_a_00713/125177/ (accessed 2026-06-15; full PDF returned HTTP 403, finding quoted from the publisher abstract/excerpt — honest gap).
- Vanilla multi-agent debate **frequently underperforms a single agent**. <https://d2jud02ci9yv69.cloudfront.net/2025-04-28-mad-159/blog/mad/> (accessed 2026-06-15); <https://arxiv.org/pdf/2502.08788> (accessed 2026-06-15).
- Cost/latency multiply per pass and per voter (a real concern for a CLI).

3. Verdict

- **(A) Wiring is sound, current best practice — YES.** TEaR is precisely this pattern and improves quality on the first refinement pass when gated by a

genuine verify/critique step. Our engine already gates (parses scores + issues, honours UNCHANGED, defaults to the existing translation), so wiring it does NOT mask a deeper problem.

- **(B) Guardrails are mandatory.** They map onto what the engine already provides + the CLI defaults we choose.

Guardrails honoured by this wiring

- **Cap passes low.** New `-multipass-passes` flag defaults to **1** (TEaR's optimum); operator may raise it but the default never regresses.
 - **Gate on a real verify signal.** The engine's per-provider LLM critique
 - consensus IS the gate (not blind "improve again").
 - **Preserve original on no improvement.** Engine defaults `PolishedText` to the existing translation and skips the `UNCHANGED` sentinel — kept intact; if `multipass` yields empty/identical text the original markdown is retained.
 - **Opt-in, back-compat preserved.** `Multipass` is a SEPARATE `-multipass` flag. The existing `-verify` heuristic path is untouched (§11.4.1 — no solve-A-create-B). `Multipass` OFF by default (it costs real LLM calls).
 - **Don't mask upstream defects.** `Multipass` runs *after* the base translation and only polishes; a base-translation failure still surfaces as a translation error before `multipass` runs.
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Sources verified 2026-06-15

- TEaR: <https://arxiv.org/html/2402.16379v3>, <https://arxiv.org/abs/2402.16379>
- QE reranking: <https://arxiv.org/html/2510.08870v1>
- MAD paradigm: <https://www.emergentmind.com/topics/multi-agent-debate-mad-paradigm>
- MAD underperformance: <https://d2jud02ci9yv69.cloudfront.net/2025-04-28-mad-159/blog/mad/>, <https://arxiv.org/pdf/2502.08788>
- MT QE metrics: <https://kvashee.medium.com/mt-quality-evaluation-in-the-age-of-llm-mt-ca8a9540a2ab>
- Self-correction limits: https://direct.mit.edu/tac1/article/doi/10.1162/tac1_a_00713/125177/ (abstract only; PDF 403 — honest gap)

Deep-research 2026-06-15: NO external solution found that contradicts wiring; TEaR confirms the pattern, guardrails enumerated above.